

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

Department of Electrical Engineering

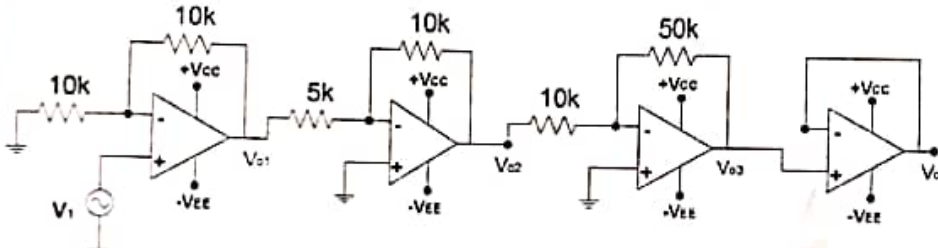
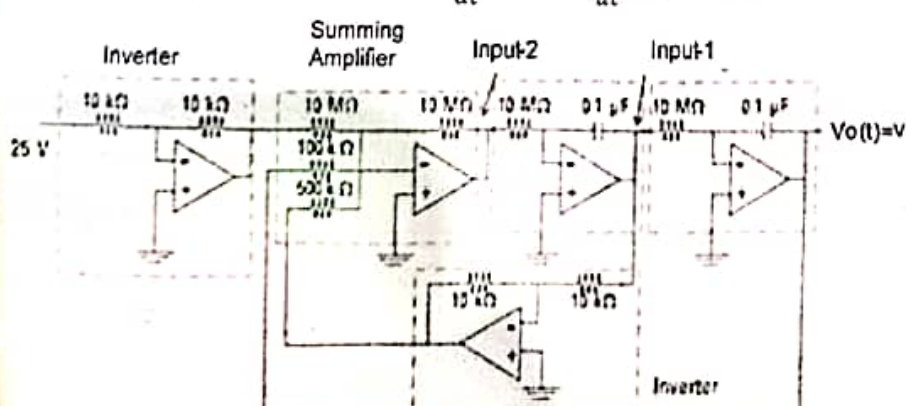
B. Tech. IV Semester Mid Term Examination March 2025

Sub: Electronics Devices & Circuits (EE-223)

Max. Marks: 200

Time: 90 Minutes

Note: Solve all questions else do as directed. Assume any missing data with due justification. Marks will be suitably scaled down.

		COs	Marks
Q1(a)	Discuss the concept of virtual ground and derive the input-output relationships of following OP-AMP Circuits (Any three): (i) Inverting Amplifier (ii) Summing Amplifier (iii) Integrator using Capacitor (iv) Differentiator using Inductor	CO2	30
(b)	Calculate the input voltage in Fig.1 below, if the final output voltage V_o is 10 V. Assume all OP AMPs are ideal and biasing voltages are $\pm 12V$. Also calculate intermediate voltages V_{o1} , V_{o2} , V_{o3} . 	CO2	30
Q.2(a)	Circuit shown in Fig.2 emulates an analog computer with output $V_o(t) = v$. From the circuit Prove that (i) Input-1 emulates $\frac{-dv}{dt}$, (ii) Input-2 emulates $\frac{d^2v}{dt^2}$ (iii) Input-2 also emulates $\frac{d^2v}{dt^2} = -20 \frac{dv}{dt} - 100v + 25$ 	CO3	40
(b)	Design a circuit that detects the zero-crossing of a periodic input signal and controls two LEDs (LED1 and LED2) to alternate their illumination based on each detected zero-crossing. The LEDs operate with a 5V HIGH (ON) and 0V LOW (OFF) logic level, and exceeding these voltage limits will damage them.	CO4	30

Q.3(a) As part of an aircraft's functional monitoring system, a circuit is required to indicate the status of the landing gears prior to landing. A green LED display turns on if all three gears are properly extended when the "gear down" switch has been activated in preparation for landing. A red LED display turns on if any of the gears fail to extend properly prior to landing. When a landing gear is extended, its sensor produces a LOW voltage. When a landing gear is retracted, its sensor produces a HIGH voltage. Implement a circuit to meet this requirement.

OR

Sensors are used to monitor the pressure and the temperature of a chemical solution stored in a vat. The circuitry for each sensor produces a HIGH voltage when a specified maximum value is exceeded. An alarm requiring a LOW voltage input must be activated when either the pressure or the temperature is excessive. Design a circuit for this application.

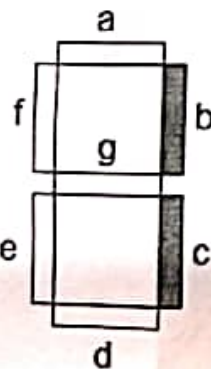
OR

In a certain automated manufacturing process, electrical components are automatically inserted in a PCB. Before the insertion tool is activated, the PCB must be properly positioned, and the component to be inserted must be in the chamber. Each of these prerequisite conditions is indicated by a HIGH voltage. The insertion tool requires a LOW voltage to activate it. Design a circuit to implement this process.

CO4

30

(b) Design a circuit for displaying decimal 1 on a seven-segment display using hexadecimal inputs. The display pattern and the corresponding truth table is given in Fig. 3. Solve the problem by using K-map and realize the circuit using minimum number of logic gates.



Inputs				Segments						
A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1
1	0	1	0	1	1	1	0	1	1	1
1	0	1	1	0	0	1	1	1	1	1
1	1	0	0	1	0	0	1	1	1	0
1	1	0	1	0	1	1	1	1	0	1
1	1	1	0	1	0	0	1	1	1	1
1	1	1	1	1	0	0	0	1	1	1

Fig. 3

CO4

40

Course Coordinator: Dr. T. S. Bheemraj

Date of Examination: Mar. 21, 2025